

Orthogonal Knowledge

Josh Pearson, Richard Roth, Ginger Schultheis · FARM · October 10, 2026

1 Orthogonal Knowledge

Sometimes our knowledge of one subject matter seems orthogonal to our knowledge of another.

- What I know about the weight of my neighbor's bicycle has nothing to do with my knowledge about how many cockatoos there are in Canberra.
- What I know about how far off my kitchen thermometer is has nothing to do with what I know about how far off my bathroom scale is.

Knowledge about two quantities may be orthogonal in three senses. In each, what you know about the first is unaffected by something about the second.

1. Whether you have measured, or even considered, the second quantity.
2. What the true value of the second is, holding your evidence fixed.
3. What evidence you have about the second.

Inquiry Orthogonality

Value Orthogonality

Evidential Orthogonality

We will investigate whether knowledge is ever orthogonal, and if so, in what sense.

2 Setup

If there is orthogonal knowledge, it will be found among quantities that are causally, evidentially, and counterfactually independent.

The thermometers. Three thermometers in isolated rooms have each taken a reading. They are identical in their chances of error: accurate with probability .7, slightly off with probability .2, wildly off with probability .1. Their errors are independent. In fact, all three are accurate. For each thermometer, there is a question.

- ① How far off is thermometer 1?
- ② How far off is thermometer 2?
- ③ How far off is thermometer 3?

Ann has only thought about ①, and has access only to the reading of the first thermometer.

Ben has only thought about ① and ②, and has access only to the readings of the first two.

Carol has thought about ①, ②, and ③, and has access to the readings of all three.

3 Five Claims

NON-TRIVIALITY. Ann knows that thermometer 1 is not wildly off.

In general: Ann knows something about ① that is not certain.

INQUIRY ORTHOGONALITY. Whatever Ann knows about 1, Ben knows, too. Whatever Ben knows about 1 and 2, Carol knows, too.

SYMMETRY. Ben knows the same about 1 and 2. Carol knows the same about 1, 2, and 3.

CLOSURE. If someone knows some things, they also know their conjunction.

THRESHOLD. Nobody knows anything whose prior probability is less than t (here, $t = .8$).

4 A Puzzle

Theorem 1. *Non-Triviality, Inquiry Orthogonality, Symmetry, Closure, and Threshold are jointly unsatisfiable.*

Proof.

- | | |
|---|--|
| (1) Ann knows 1 is not wildly off. | (Non-Triviality) |
| (2) Ben knows 1 is not wildly off. | (IO, $\text{Ann} \rightarrow \text{Ben}$) |
| (3) Ben knows 2 is not wildly off. | (Symmetry, Ben) |
| (4) Carol knows 1 and 2 are not wildly off. | (IO, $\text{Ben} \rightarrow \text{Carol}$) |
| (5) Carol knows 3 is not wildly off. | (Symmetry, Carol) |
| (6) Carol knows none of the three is wildly off. | (Closure) |
| (7) $P(\text{none wildly off}) = .9^3 \approx .73 < .8$. | (contradicts Threshold) $\square$ |

Three thermometers suffice for $t = .8$. For any $t < 1$, enough thermometers will do. A view that lowers t as n grows escapes every finite case. To refute such a view, we need infinitely many thermometers.

5 Five Ways Out

Deny Non-Triviality.

Too much skepticism.

Deny Threshold.

Variant of puzzle with chance.

Deny Closure.

We set this aside for this talk.

Deny Inquiry Orthogonality. Carol knows less about instrument 1 than Ann.

We explore this in §6.

Deny Symmetry.

We explore this in §7.

6 Denying Inquiry Orthogonality

The View. Carol knows less than Ann does. Ann can rule out an answer to ①. Carol cannot rule out any answer to ①, ②, or ③.

Why? The more questions you have considered, the harder it is to satisfy Threshold, and so the harder it is for you to know.

A Question. What kinds of evidence can give Carol Ann’s knowledge about ①?

- (a) What if Carol learns the exact answers to ② and ③?
- (b) What if Ann tells Carol what she knows?

A Puzzle about (a). Let Diana be exactly like Carol except that she has learned the exact answers to ② and ③. Two principles give opposite answers about what Diana knows about ①.

EVIDENTIAL ORTHOGONALITY

Learning the true temperatures of Rooms 2 and 3 doesn’t change what Diana knows about ①. What Diana knows on the basis of thermometer 1 is the same as what Carol knows.

DIMENSIONAL REDUCTION

Once you know the exact answers to ② and ③, you are in the position of someone with one thermometer. What Diana knows on the basis of thermometer 1 is the same as what Ann knows.

Theorem 2. *Non-Triviality, Threshold, Closure, Symmetry, Evidential Orthogonality, and Dimensional Reduction are jointly unsatisfiable.*

A puzzle about (b). Two possibilities:

- Carol learns that Thermometer 1 is not wildly off. That’s weird: Ann’s testimony gives Carol no evidence about the thermometers that she didn’t already have.
- Carol does not learn that Thermometer 1 is not wildly off. Also weird: testimony from less sophisticated knowers will fail to yield knowledge all the time.

Ruling out an answer to ① commits Carol, by Symmetry, to ruling out an answer to ② and ③ too, and by Closure to their conjunction, which Threshold forbids.

We are inclined to take the DR horn. IO denier’s motivation: more uncertain questions, more risk, less knowledge. Instruments whose values you know contribute no risk.

Proof Sketch. By EO, Diana knows what Carol knows about ①. By DR, Diana knows what Ann knows. So what Carol knows about ① is what Ann knows about ①. The same argument, run for any question and any two people, yields IO in full. Now apply Theorem 1.

7 Denying Symmetry: A Counterfactual Theory

The View. Carol knows different things about ①, ②, and ③.

Seen from many Anns. For each instrument, there is a specialist—call her Ann_i —who holds only instrument i . Carol holds all of them. By IO (generalized), Carol knows about i everything that Ann_i knows about i . So not all Anns know that their instruments are not wildly off (Proposition 1).

Proposition 1. *Threshold, Closure, and Inquiry Orthogonality imply that at most k Anns know that their instrument is not wildly off where k is the largest number with $0.9^k \geq t$ (here, 2).*

Question. If we keep Threshold, Closure, and IO, Symmetry fails. Some Anns know, and others don't. Why the difference?

Rough Answer. Not a fact about probability. Instead: a counterfactual fact. Whoever would too easily have been mistaken fails to know.

7.1 Emeralds

1000 emeralds. Either it's a law that all are green, or each is green or blue at random. In fact it's a law.

Puzzle due to Andrew Bacon (2020).

Four Principles.

- *Inductive Knowledge* (IK). Seeing 100 green emeralds, you know it's a law.
- *Anonymity*. Any 100 emeralds are as good as any other.
- *What You See Is What You Know* (WYSIWYK). Each observation only tells you that this one is green.
- *Closure*.

The Problem. Together, the four principles give you too much knowledge.

1. Before looking you knew: if these 100 are green, it's a law. (IK, WYSIWYK)
2. Same for every group of 100. (Anonymity)
3. So before looking you knew: for every 100, if all green, it's a law. (Closure)
4. Contrapose: if it's not a law, more than 900 are blue. Absurd.

Bacon's Escape. Deny Anonymity. A group of 100 emeralds gives law knowledge only if enough of them would have been blue had there been no law.

See also Williamson (2000, 2009) and Sosa (1999) for related views.

A Variant. Drop the 'no law' antecedent. Order worlds by how easily they could have obtained, actual world first. You can't rule out the first m worlds, where m is the smallest number making them jointly t -likely.

The Lottery. 1000 tickets. Ticket 17 wins. Had 17 not won, 42 would have. Had neither 17 nor 42 won, 8 would have. Order worlds that way: 17, 42, 8, and so on. You can't rule out the first m , the fewest that are jointly t -likely. Of those m , the $m-1$ losers don't know they lost. The other losers do.

7.2 A Parallel Escape

Whether Ann_i can rule out wild error on her instrument is settled by a counterfactual fact: whether she would too easily have been mistaken.

The Construction. As in the variant, with worlds now assignments to all three thermometers. $S_t(w)$ is the first m worlds, m the smallest number making them jointly t -likely. You know what's true at every evidence-compatible world in $S_t(w)$.

The Anns. The actual world is 000. Suppose the ordering runs:

000 < 100 < 010 < 001 < 002 < 110 < 101 < 011 < 102 < 012 < 111 ...

First ten worlds have probability .798. First eleven .806. So $S_t(w)$ is the first eleven. Thermometer 3 is wildly off at three of them, 1 and 2 at none. Ann_1 and Ann_2 know their instruments are not wild. Ann_3 doesn't.

What holds. Threshold. (The open worlds are jointly t -likely.) Inquiry Orthogonality. (Carol knows about i exactly what Ann_i knows.) Non-Triviality. (For Anns whose thermometers fail only after first m worlds.) Closure.

Drawbacks.

- Value Orthogonality fails. (See next section.)
- Infinity Trouble: It can happen that every initial segment of the counterfactual ordering is either probability 1, 0 or unmeasurable.

8 Value Orthogonality and The Counterfactual Theory

Value Orthogonality (VO). What Carol knows about thermometer j does not depend on how far off the other thermometers actually are, holding her evidence fixed.

The Question. The counterfactual theory gets IO by giving up Symmetry. Can it keep VO? No. Two natural assumptions bring Symmetry back.

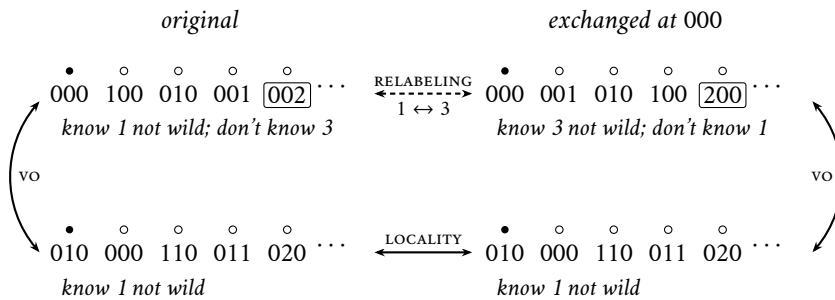
Supervenience. Two parts.

Locality. What you know at a world depends only on the probabilities and the ordering *at that world*.

Relabeling. Renaming the thermometers renames what is known, nothing more. Exchange 1 and 3 throughout the ordering at that world, and what you knew about 1 you now know about 3.

Richness. Any ordering at a world could be replaced by the same ordering with two thermometers exchanged, leaving the ordering everywhere else the same.

Proposition 2. *Supervenience, Richness, and Value Orthogonality imply Symmetry for thermometers with the same chances.*



Relabeling and Locality fix every row but the top right. VO then forces the two 000 rows to agree about 1. But on the left you know 1 isn't wild and the right you don't.

The Lesson. Two sorts of theories, two sorts of orthogonality.

- *Probabilistic theories.* Same probabilities, same knowledge: Symmetry holds. By Theorem 1, Inquiry Orthogonality fails.
- *Counterfactual theories.* Knowledge is fixed by what could easily have happened, not by your questions: Inquiry Orthogonality can hold. But then Symmetry fails, and so by Proposition 2 Value Orthogonality fails too.

Knowledge can be orthogonal, but not in every sense at once. Whether it answers to probability or to counterfactual danger decides which we can keep.

E.g. countably many thermometers, countable additivity, the Continuum Hypothesis, and an ordering of type ω_1 . Every proper initial segment is countable, so has probability 0. No initial segment short of W itself reaches t .

Formally: For all $\langle W, P, <, R \rangle$, $\langle W, P', <', R' \rangle \in \mathcal{C}$, any coordinate swap τ , and any $w \in W$, if $P'(\tau v) = P(v)$ for all $v \in W$ and $u <_w v$ iff $\tau u <_{\tau w} \tau v$ for all $u, v \in W$, then $R'(\tau w) = \tau R(w)$.

FIGURE 1. Each row is the counterfactual ordering at a world: the world itself, then the next four. The boxed world is the one that makes you lose knowledge. Solid arrows: same knowledge about ①. Dashed arrow: knowledge about ① and ③ swapped. VO is violated on the right.

Value Orthogonality is optional. Goodman & Salow violate it. But other theories in their vein validate it.

The other values shape the ordering, and so where the cut falls.